

High-dimensional nonattainment for spatially weighted L^2 -BV regularization

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Abstract

Theorem 2.8 of Mazziere–Spies–Temperini asserts existence and uniqueness of a minimizer for a mixed spatially weighted L^2 -BV functional in every dimension under the assumptions $(1 - \theta)^{-1} \in L^1$ and $\theta^{-1} \in L^\infty$. We give an explicit counterexample in every dimension $n \geq 3$. A sequence of shrinking BV spikes fits the datum exactly while both penalties tend to zero, so the infimum is zero and is not attained. We also prove the sharp dimensional repair: the stated hypotheses do give existence and uniqueness for $n \leq 2$. The proof needs only the continuous critical embedding $BV(\Omega) \hookrightarrow L^2(\Omega)$, not the compact embedding claimed in the source.

1 The source claim and the dimension gap

Let $\Omega \subset \mathbb{R}^n$ be bounded with Lipschitz boundary and let $0 \leq \theta \leq 1$. The source defines

$$W_{0,\theta}(u) = \sup_{\nu \in \mathcal{V}_\theta} \int_{\Omega} -u \operatorname{div}(\theta \nu) dx$$

and studies, for $\alpha_1, \alpha_2 > 0$,

$$F_\theta(u) = \|Tu - v\|_Y^2 + \alpha_1 \int_{\Omega} (1 - \theta)|u|^2 dx + \alpha_2 W_{0,\theta}(u), \quad u \in L^2(\Omega). \quad (1)$$

Theorem 2.8 asserts that if

$$\frac{1}{1 - \theta} \in L^1(\Omega), \quad \frac{1}{\theta} \in L^\infty(\Omega), \quad (2)$$

then (1) has a unique global minimizer in $BV(\Omega)$, with no restriction on n [1]. Its proof obtains a BV bound and then invokes a “compact embedding of $BV(\Omega)$ in $L^2(\Omega)$.” Such an embedding cannot supply an L^2 bound when $n \geq 3$.

We use two elementary facts already available in the source:

$$0 \leq W_{0,\theta}(u) \leq J_0(u), \quad J_0(u) \leq \|\theta^{-1}\|_\infty W_{0,\theta}(u) \quad \text{if } \theta^{-1} \in L^\infty. \quad (3)$$

2 Counterexample in every dimension at least three

Theorem 1. *For every integer $n \geq 3$ there are a ball $\Omega \subset \mathbb{R}^n$, a weight θ satisfying (2), a bounded linear map $T : L^2(\Omega) \rightarrow \mathbb{R}$, and $v \in \mathbb{R}$ such that F_θ has infimum zero but no global minimizer. Thus the existence assertion in source Theorem 2.8 is false in every dimension $n \geq 3$.*

Proof. Fix $0 < R < 1$, put $\Omega = B(0, R)$, and set

$$\theta(x) = 1 - |x|^{n-1}, \quad \gamma = \frac{n+2}{4}, \quad g(x) = |x|^{-\gamma}.$$

Because $n \geq 3$,

$$1 < \gamma < \frac{n}{2}.$$

Consequently $g \in L^2(\Omega)$, and

$$Tu := \int_{\Omega} g(x)u(x) dx$$

defines a bounded linear functional on $L^2(\Omega)$. Take $v = 1$. The weight has

$$\theta(x) \geq 1 - R^{n-1} > 0, \quad \frac{1}{1 - \theta(x)} = |x|^{-(n-1)} \in L^1(\Omega),$$

so both source hypotheses hold.

Let $\sigma_{n-1} = |S^{n-1}|$. For $0 < r < R$, define

$$c_r = \frac{n - \gamma}{\sigma_{n-1}} r^{\gamma-n}, \quad u_r = c_r \mathbf{1}_{B(0,r)}.$$

Each u_r belongs to $L^2(\Omega) \cap \text{BV}(\Omega)$, and radial integration gives

$$Tu_r = c_r \sigma_{n-1} \int_0^r \rho^{n-1-\gamma} d\rho = 1. \quad (4)$$

Thus the data term in (1) vanishes identically along this sequence. The weighted quadratic term is

$$\begin{aligned} \int_{\Omega} (1 - \theta) |u_r|^2 dx &= c_r^2 \sigma_{n-1} \int_0^r \rho^{2n-2} d\rho \\ &= \frac{(n - \gamma)^2}{\sigma_{n-1}(2n - 1)} r^{2\gamma-1} = \frac{(n - \gamma)^2}{\sigma_{n-1}(2n - 1)} r^{n/2}. \end{aligned} \quad (5)$$

Moreover, by (3) and the perimeter formula for a ball,

$$W_{0,\theta}(u_r) \leq J_0(u_r) = c_r \sigma_{n-1} r^{n-1} = (n - \gamma) r^{\gamma-1} = (n - \gamma) r^{(n-2)/4}. \quad (6)$$

Both exponents in (5)–(6) are positive. Therefore

$$0 \leq F_{\theta}(u_r) \leq \alpha_1 \frac{(n - \gamma)^2}{\sigma_{n-1}(2n - 1)} r^{n/2} + \alpha_2 (n - \gamma) r^{(n-2)/4} \longrightarrow 0.$$

Hence $\inf_{L^2} F_{\theta} = 0$.

If a minimizer u_* existed, all three nonnegative terms in (1) would have to vanish. Since $1 - \theta = |x|^{n-1} > 0$ almost everywhere, vanishing of the weighted quadratic term forces $u_* = 0$ almost everywhere. But then $Tu_* = 0 \neq 1 = v$, contradicting vanishing of the data term. Thus the infimum is not attained. $\square$

Remark 1. Taking the zero set $\{\theta = 0\}$ to be empty shows that the same example also contradicts the unrestricted-dimensional existence assertion in source Theorem 2.10, whose hypotheses then specialize to those of Theorem 2.8.

3 The sharp repair in dimensions one and two

Proposition 1. *Under the source hypotheses (2), the conclusion of source Theorem 2.8 is valid when $n \leq 2$: the functional F_θ has a unique global minimizer in $BV(\Omega)$.*

Proof. Consider a sequence (u_k) on which the two penalty terms are bounded. Cauchy–Schwarz and $(1 - \theta)^{-1} \in L^1$ give

$$\|u_k\|_{L^1} \leq \left\| \frac{1}{1 - \theta} \right\|_{L^1}^{1/2} \|\sqrt{1 - \theta} u_k\|_{L^2}.$$

The second inequality in (3) gives a uniform bound on $J_0(u_k)$. Hence (u_k) is bounded in $BV(\Omega)$. For $n = 2$, the critical Sobolev inequality gives the continuous embedding $BV(\Omega) \hookrightarrow L^2(\Omega)$; for $n = 1$, BV embeds into L^∞ , hence also into L^2 . Thus (u_k) is bounded in the reflexive space $L^2(\Omega)$ and has a weakly convergent subsequence.

The source proves weak lower semicontinuity of $W_{0,\theta}$; the other terms are weakly lower semicontinuous as well. The direct method therefore produces a minimizer. The reciprocal assumption implies $1 - \theta > 0$ almost everywhere, so $u \mapsto \|\sqrt{1 - \theta} u\|_2^2$ is strictly convex. This gives uniqueness, while the displayed L^1 and variation bounds put the minimizer in $BV(\Omega)$. $\square$

Proof Intuition

The condition $(1 - \theta)^{-1} \in L^1$ controls the L^1 mass of a candidate, and the weighted variation controls its BV seminorm. In dimensions one and two, those two controls automatically bound the L^2 norm, so weak L^2 compactness completes the direct method. In dimension three and higher, BV is subcritical for L^2 : a shrinking ball can carry taller and taller spikes while its perimeter cost vanishes. By placing the degenerating quadratic weight at the same point and choosing a singular but square-integrable observation functional, the spikes fit the datum exactly. The objective approaches zero, although zero cost itself would force the candidate to be both 0 and data-fitting. This is the precise compactness failure hidden by the dimension-free statement.

References

- [1] G. L. Mazzieri, R. D. Spies, and K. G. Temperini, *Mixed spatially varying L^2 – BV regularization of inverse ill-posed problems*, arXiv:1403.5579 (2014), especially Theorem 2.8 and its proof on pp. 7–8.